Ruchir Barhate

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Education

Indian Institute of Technology Bombay — Mumbai, India

B.Tech, Industrial Engineering & Operations Research; Minors in Statistics and AI & DS — *Expected 2028*

CGPA: 9.73/10 | **Department Rank 2** in Industrial Engineering & Operations Research (2026)

- Ranked **Institute Rank 6** out of 1400+ students across all branches at IIT Bombay, by CGPA (2025)
- Secured **All India Rank 1720** out of 2.25 lakh+ candidates in the JEE Advanced examination (2024)
- Selected as Teaching Assistant for Probability, Calculus and Optimization courses in 3 semesters (2025-26)
- Earned **19 AA(10/10)** and 5 AB(9/10) grades over first two years, reflecting consistent performance (2026)
- Won the Computer Programming and Utilisation lab contest, placing in the **top 4%** of 650+ students (2025)

Experience

CSALT (Computational Speech and Language Technologies), IIT Bombay — Mumbai, India

Multilingual LLM Proficiency (Prof. Preethi Jyothi) — *Aug 2026 – Present*

Researching how to fairly measure how well large language models understand Indian languages, beyond perplexity

- Showed perplexity unfairly favours languages split into more tokens, rating Telugu (**27.3**) easier than Hindi (**44.9**)
- Modelled attention as a token graph and tracked its spectral structure across all **48 layers** of Gemma in **8** languages
- Found graph metrics mostly tracked text length ($\mathbf{r = 0.98}$), so designed a fairer test on **100** parallel texts per language

TCAAI, IIT Bombay — Mumbai, India

Dialect Classification (Prof. Manjesh Hanawal) — *Mar 2026 – Aug 2026*

Developed a robust speech classification architecture deploying state-of-the-art transformer networks for dialects

- Assembled a **200+ hour** audio corpus by deploying a **4-agent** workflow to autonomously search, verify and download
- Extracted dialect-discriminative signals by attaching a **25-scalar** WeightedLayerPool to the WavLM Large backbone
- Preserved core phonetic structures by selectively applying LoRA adapters across the **24-layer** transformer hierarchy

Key Projects

BLADE: BNN Layer-wise Activation Driven Engine (Course Project) — *Jan 2026 – Apr 2026*

Designed BLADE, a decomposition framework for Binarized Neural Network training (Prof. Avinash B, Prof. Vishnu N)

- Reduced search space by **92 orders of magnitude** by reformulating weight search as set-cover pattern selection
- Achieved a **4x speedup** over monolithic HA while matching **91.67%** test accuracy vs gradient descent's 86.67%
- Generated **5,000** activation patterns via random sampling, structured perturbation, and correlation on labels

Federated CNNs for Plant Disease Classification — *Dec 2025 – Jan 2026*

Developed a high-precision disease classification system using deep learning ensembles on the PlantVillage dataset

- Engineered a **3-block CNN** from scratch with batch norm and dropout, reaching **95.5%** accuracy on 38 classes
- Fine-tuned a pretrained ResNet18 on **54,305 images** in two stages, lifting validation accuracy to **99.6%**
- Trained ResNet18 across **3 isolated clients** via FedAvg in Flower, cutting loss **8.8x** with zero raw-data sharing

International Competitions

International Rocket Engineering Competition (IREC) 2026 — Midland, Texas, USA

IIT Bombay Rocket Team, Avionics Subsystem (Project Ananta) — *Oct 2024 – Aug 2026*

Represented IIT Bombay among 180+ universities worldwide, building avionics for a 10,000 ft, 0.9 Mach SRAD rocket

- Secured **Category Rank 3**, becoming the first team from Asia to successfully launch a **10,000 ft** SRAD rocket
- Built a Mission Control GUI using HTML, CSS & JS to decode LoRa packets and visualize live flight telemetry
- Designed **4-layer PCBs** in KiCad for flight state detection, autonomous parachute deployment, and RF telemetry

Leadership & Activities

IEOR Undergraduate Representative — 2026 – 2027

- Appointed as **Undergraduate Representative** for IEOR, serving as the primary liaison to faculty

Hindustani Classical Music & Tabla — 2017 – 2024

- Completed 7+ years of training in Tabla and Classical Music, earning **three Distinctions** and a First Class
- Performed for commencement and interim convocation to an audience of 1400+ graduating students
- Performed a solo (Abhang geet) and group performance for the "Swachhata Hi Seva" event on 2nd Oct

Skills & Interests

Technical: C++, Python, Jupyter Notebook, Gurobi, MicroPython, MySQL, PyTorch, TensorFlow, Scikit-learn
Courses: Data Structures & Algorithms, Derivative Pricing, Non-linear & Discrete Optimization
Certifications: Completed the Deep Learning NLP and CV Bootcamp offered by the Analytics Club, IIT Bombay
Interests: Tabla, Hindustani Classical Music, Optimization, AI & ML, Quantitative Finance, Probability